SMITH ENGINEERING

Queen's University

Department of Electrical and Computer Engineering



ELEC 490/498 Project Blueprint

Autonomous Smart Caddy Cart (ASCC)

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Acronyms

1. Introduction

1.1 Motivation and Problem

Golfers who walk the course must transport their equipment by carrying a bag or operating a push cart. Although electric and autonomous carts reduce this effort, they primarily focus on transportation rather than equipment management. Golfers must also keep track of clubs removed during play, which can be misplaced or left behind.

The Autonomous Smart Caddy Cart (ASCC) aims to address both challenges by combining autonomous golfer following with club identification and tracking. As golfers ourselves, the team understands the need for a solution that addresses both transportation and equipment management. The goal is to reduce manual cart operation while helping golfers manage their equipment throughout a round.

1.2 Project Objectives

The ASCC will be developed as a motorized golf push cart capable of following a golfer using a controller-based tracking system rather than computer vision. Distance sensors will support obstacle detection and avoidance, while an emergency-stop mechanism will allow the cart to be stopped when necessary.

An RFID-based subsystem will identify tagged golf clubs and monitor their presence in the bag, notifying the golfer when a club is missing. The objective is to integrate these functions into a reliable and safe prototype.

The ASCC will also aim to incorporate golf course data, including hole layouts, distances, and course information, to assist golfers with shot planning. This information helps golfers estimate distances to targets and hazards, supporting more informed club selection and on-course decisions.

1.3 Project Scope

The minimum viable prototype will focus on motor control, golfer following, obstacle detection, emergency stopping, RFID-based club tracking, and basic golf course data integration. Additional features, including shot data collection, club recommendations, and speech functionalities, may be implemented if time and resources permit.

Testing will begin indoors before progressing to controlled outdoor environments. The prototype is not intended for commercial deployment or operation on public roads, crowded areas, or extreme terrain.

1.4 System Overview and Operating Scenario

Figure 1 illustrates the ASCC's main subsystems and their interactions with the golfer and surrounding environment. The Raspberry Pi coordinates sensor inputs, motor control, and club tracking.

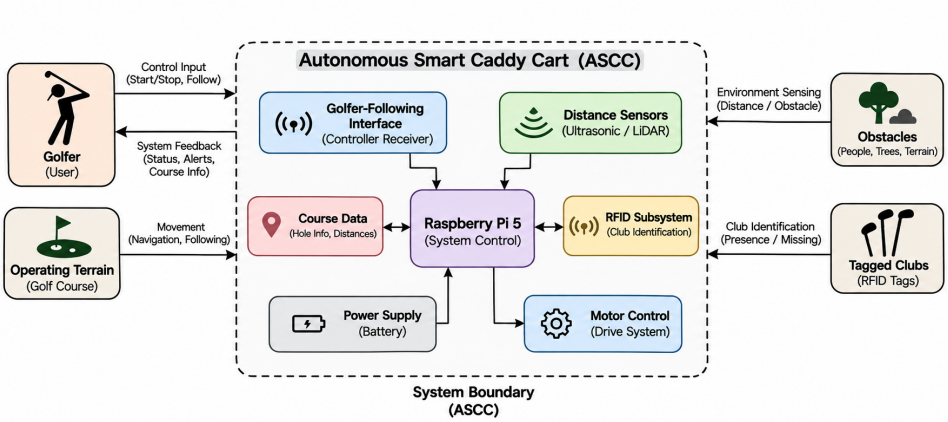


Figure 1: ASCC system overview and external interactions.

1.5 Design Constraints

The ASCC must be completed within the CAD $600 budget and capstone development schedule. Existing components, including the Raspberry Pi, ultrasonic sensors, and golf push cart, will be reused to reduce costs.

Key constraints include hardware compatibility, sensor reliability, battery capacity, and safe motor operation. These considerations will guide the system requirements, development approach, and verification procedures.

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7. Resource Requirements

The ASCC has a project budget of CAD $600. Several key components, including the Raspberry Pi, ultrasonic sensors, and golf push cart, are already owned or borrowed, reducing the overall cost. The remaining budget provides flexibility for unexpected expenses, replacements, or design changes. Additional components, including motors and GPS tracking hardware, may be obtained through the ECE Department or Ingenuity Labs. However, their estimated costs are included in the budget until their availability is confirmed. Table 1 summarizes the estimated costs.

7.1 Hardware and Equipment

Table 1: Estimated ASCC bill of materials (CAD).

Component	Qty.	Cost
Raspberry Pi 5 (existing)	1	$0
Raspberry Pi display	1	$40
Ultrasonic sensors (existing)	Multiple	$0
RFID readers and tags	2 sets	$20
TF-Luna LiDAR sensor	1	$50
GPS tracker	1	$30
Golf push cart (existing)	1	$0
Motors	1 set	$80
Motor controller	1	$50
Battery and power supply	1 set	$75
Mounting hardware	Assorted	$30
Wiring, connectors, switches	Assorted	$30
Tax and shipping	–	$75
Total		**$480**

7.2 Facilities and Contingencies

Development will require electronics tools and access to indoor and outdoor testing areas. Tools can be found utilizing the ’65 ECE lab, and testing will occur in controlled areas and hopefully on a real golf course for the finished prototype. Python and Git will support software development.

The estimated cost is $480, leaving $120 below the $600 budget limit. This remaining budget allows for additional components or design adjustments if needed.

8. Conclusion

The ASCC blueprint builds upon the initial proposal by establishing a more detailed system design, measurable performance requirements, and contingency plans for potential failures. The project continues to focus on autonomous golfer following, obstacle de-

tection, emergency stopping, RFID-based club tracking, and golf course information to improve the walking golf experience.

A key focus of the blueprint is justifying quantitative performance targets based on hardware capabilities, operating conditions, and safety considerations. These targets provide measurable acceptance criteria for verification and validation testing, ensuring that system performance can be evaluated against clearly defined requirements. The incremental development approach also allows individual prototypes to be tested before integration, reducing the risk of unresolved subsystem failures.

Potential limitations include unreliable sensor readings, inconsistent RFID detection, loss of golfer tracking, and difficulties operating on uneven terrain. The blueprint addresses these risks through alternative hardware configurations, adjustments to sensor placement, controlled testing, and fail-safe stopping behaviour. If technical or scheduling constraints arise, optional features will be deferred to prioritize safe and reliable operation of the core system.

With an estimated budget of CAD $480 and access to existing equipment, the project remains feasible within the $600 limit. The team has established its system architecture and identified the major technical requirements and resources. The next steps involve finalizing quantitative acceptance criteria, confirming hardware availability, and developing subsystem prototypes to validate the proposed design before full integration.

9. References

A. Detailed Requirements and Traceability

B. Supporting Technical Calculations

C. Detailed Test Procedures

D. Detailed Resource and Budget Information